

Online Learning

Other topics

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Evaluation of Bandit Algorithms

Evaluation of bandit algorithms in practice

- Challenge: previously unobserved actions or (state,action) pairs
- Deployment
 - Risky and time-consuming
- Environment simulation
 - Requires a good simulator
 - This may be very hard or even impossible to produce
 - If we have a good simulator, we probably already have a solution to the problem

Evaluation of bandit algorithms in practice

- Offline evaluation for i.i.d. problems
 1. Use full information data where possible and relevant
 2. “Importance-weighting” of logged limited feedback data
 - Requires randomized sampling in the logging policy with non-zero probability for taking all the (potentially relevant) actions
 - Requires logging the sampling distribution (to do importance-weighting)
 - Variance of the estimates scales with $\frac{1}{p_{\text{logging}}(a)}$
 - So for every potentially relevant action $p_{\text{logging}}(a)$ should not be too small
 - You will experiment with this approach in the home assignment!
- Evaluation in the adversarial regime
 - Generally impossible
 - Sparring

Alternative algorithms for bandits

Alternative algorithms for i.i.d. bandits

- UCB-style algorithms
 - kl-UCB (based on kl inequality)
 - UCB-V (based on Empirical Bernstein or Unexpected Bernstein inequality)
- Thompson sampling (Bayesian approach)
- Subsampling
- Best-of-both-worlds algorithms

Variations of EXP3 – high probability regret bound

- EXP3

- $p_t(a) = \frac{e^{-\eta_t L_{t-1}(a)}}{\sum_{a'} e^{-\eta_t L_{t-1}(a')}}$
 - $\tilde{\ell}_{t,a} = \frac{\ell_{t,a} \mathbb{I}(A_t=a)}{p_t(a)}$
 - $\mathbb{E}[R_T] = O(\sqrt{KT \ln K})$

- EXP3-IX: high-probability regret guarantee

- $\tilde{\ell}_{t,a} = \frac{\ell_{t,a} \mathbb{I}(A_t=a)}{p_t(a) + \frac{\eta_t}{2}}$
 - $\mathbb{P}\left(R_T \geq O\left(\sqrt{KT \ln K \ln \frac{1}{\delta}}\right)\right) \leq \delta$

Variations of EXP3 – best-of-both-worlds

$$\sum_{t=1}^T \frac{1}{\sqrt{t}} \leq 2\sqrt{T}$$
$$\sum_{t=1}^T \frac{1}{t} \leq 1 + \ln T$$
$$\sum_{t=1}^T \frac{1}{t\Delta(a)^2} \leq \frac{1 + \ln T}{\Delta(a)^2}$$

- EXP3

- $p_t(a) = \frac{e^{-\eta_t L_{t-1}(a)}}{\sum_{a'} e^{-\eta_t L_{t-1}(a')}$
 - $\tilde{\ell}_{t,a} = \frac{\ell_{t,a} \mathbb{I}(A_t=a)}{p_t(a)}$
 - $\mathbb{E}[R_T] = O(\sqrt{KT \ln K})$

- EXP3++: best-of-both-worlds

- $\tilde{p}_t(a) = (1 - \sum_a \varepsilon_t(a))p_t(a) + \varepsilon_t(a)$
 - $\varepsilon_t(a) = \theta \left(\frac{\ln t}{t \hat{\Delta}_t(a)^2} \right)$, where $\hat{\Delta}_t(a)$ is a lower confidence bound on the gap
 - $\mathbb{E}[R_T] = O(\sqrt{KT \ln K})$
 - $\bar{R}_T = O\left(\sum_{a:\Delta(a)>0} \frac{(\ln T)^2}{\Delta(a)}\right)$

Adversarial bandits: alternative regularization

- EXP3

- $p_t = \frac{e^{-\eta_t L_{t-1}(a)}}{\sum_{a'} e^{-\eta_t L_{t-1}(a')}} = \arg \min_p \langle p, L_{t-1} \rangle + \underbrace{\frac{1}{\eta_t} \sum_a p_a \ln p_a}_{\substack{\text{Regularization} \\ \text{Negative entropy}}}$

- Tsallis-INF – the ultimate algorithm: Best-of-both-worlds and minimax optimal

- $p_t = \arg \min_p \langle p, L_{t-1} \rangle - \underbrace{\frac{1}{\eta_t} \sum_a \sqrt{p_a}}_{\substack{\text{Regularization} \\ \text{Tsallis entropy}}}$

- Adversarial: $\mathbb{E}[R_T] = O(\sqrt{KT})$
 - I.I.D.: $\bar{R}_T = O\left(\sum_{a: \Delta(a) > 0} \frac{\ln T}{\Delta(a)}\right)$

Tsallis-INF

[Zimmert & Seldin, JMLR 2021; Masoudian & Seldin, COLT 2021; Ito, COLT 2021]

Tsallis-INF Algorithm:

- $\forall a: \hat{L}_0(a) = 0$
- For $t = 1, 2, \dots$
 - $\forall a: p_t(a) = \arg \min_{p \in \Delta^{K-1}} \left(\langle \hat{L}_t, p \rangle - \frac{1}{\eta_t} \sum_{a'} \sqrt{p(a')} \right)$
 - $A_t \sim p_t$
 - $\forall a: \hat{L}_t(a) = \hat{L}_{t-1}(a) + \frac{\ell_{t,a} \mathbb{1}(A_t=a)}{p_t(a)}$
- Original analysis [Audibert & Bubeck, COLT 2021]:

$$\bar{R}_T \leq c \sum_{t=1}^T \frac{1}{\sqrt{t}} \mathbb{E} \left[\sum_a \sqrt{p_t(a)} \right] = O(\sqrt{KT})$$

- Observation: $p_t(a^*) = 1 - \sum_{a \neq a^*} p_t(a)$

$$\bar{R}_T \leq c \sum_t^T \frac{1}{\sqrt{t}} \mathbb{E} \left[\sum_{a \neq a^*} \sqrt{p_t(a)} \right] = O(\sqrt{KT})$$

- Self-bounding analysis:

If $\bar{R}_T = \sum_t^T \sum_a \Delta(a) \mathbb{E}[p_t(a)]$ then

$$\begin{aligned} \bar{R}_T &= 2\bar{R}_T - \sum_{t=1}^T \sum_{a \neq a^*} \Delta(a) \mathbb{E}[p_t(a)] \\ &\leq \sum_t^T \mathbb{E} \left[\sum_{a \neq a^*} \frac{2c\sqrt{p_t(a)}}{\sqrt{t}} - \Delta(a)p_t(a) \right] \\ &\leq \sum_t^T \sum_{a \neq a^*} \frac{c'}{t\Delta(a)} = O \left(\sum_{a \neq a^*} \frac{\ln T}{\Delta(a)} \right) \end{aligned}$$

Tsallis-INF – summary

[Zimmert & Seldin, JMLR 2021; Masoudian & Seldin, COLT 2021; Ito, COLT 2021]

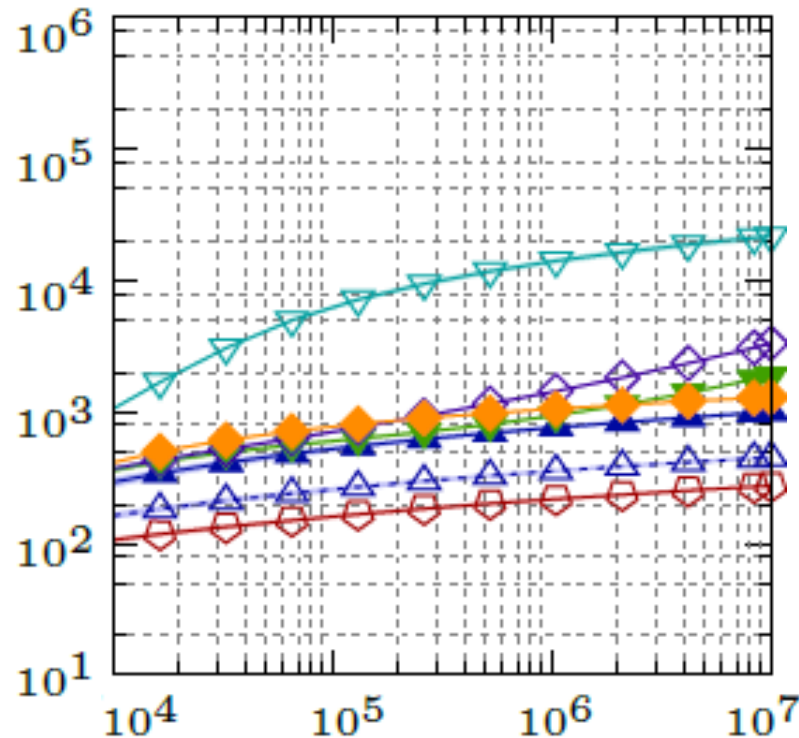
- Adversarial: $\bar{R}_T = O(\sqrt{KT})$
- Stochastic: $\bar{R}_T = O\left(\sum_{a \neq a^*} \frac{\ln T}{\Delta(a)}\right)$
- Bonus: refined regret guarantees in adversarial regimes with (Δ, C, T) self-bounding constraints $\bar{R}_T \geq \sum_t^T \sum_a \Delta(a) \mathbb{E}[p_t(a)] - C$
 - Stochastically constrained adversarial bandits:
 - The gaps $\Delta(a)$ are fixed, but the means $\mu(a)$ can fluctuate
 - $\bar{R}_T = \sum_t^T \sum_a \Delta(a) \mathbb{E}[p_t(a)]$ (the same as in the stochastic bandits)
 - $\bar{R}_T = O\left(\sum_{a \neq a^*} \frac{\ln T}{\Delta(a)}\right)$ (the same as in the stochastic bandits)
 - Stochastic bandits with adversarial corruptions: $\bar{R}_T = O\left(\sum_{a \neq a^*} \frac{\ln T}{\Delta(a)} + \sqrt{C \sum_{a \neq a^*} \frac{\ln T}{\Delta(a)}}\right)$

Tsallis-INF in practice

[Zimmert & Seldin, JMLR, 2021]

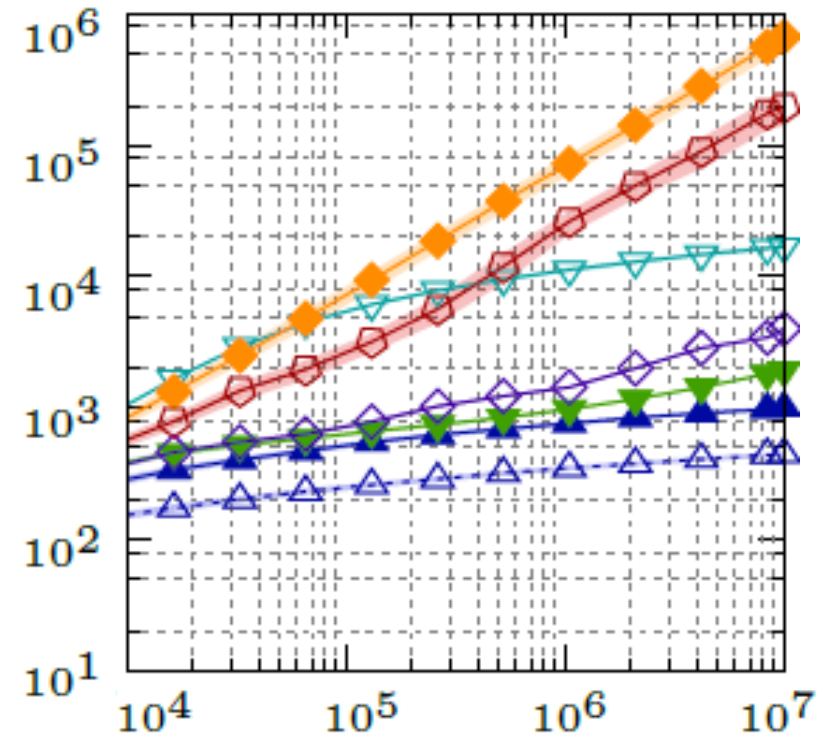
$$K = 8, \Delta = 0.125$$

T-INF RV --- △
T-INF IW — ▲
BROAD — ▽
EXP3++ — ▼
EXP3 — ◇
UCB1 — ◆
TS — ◻



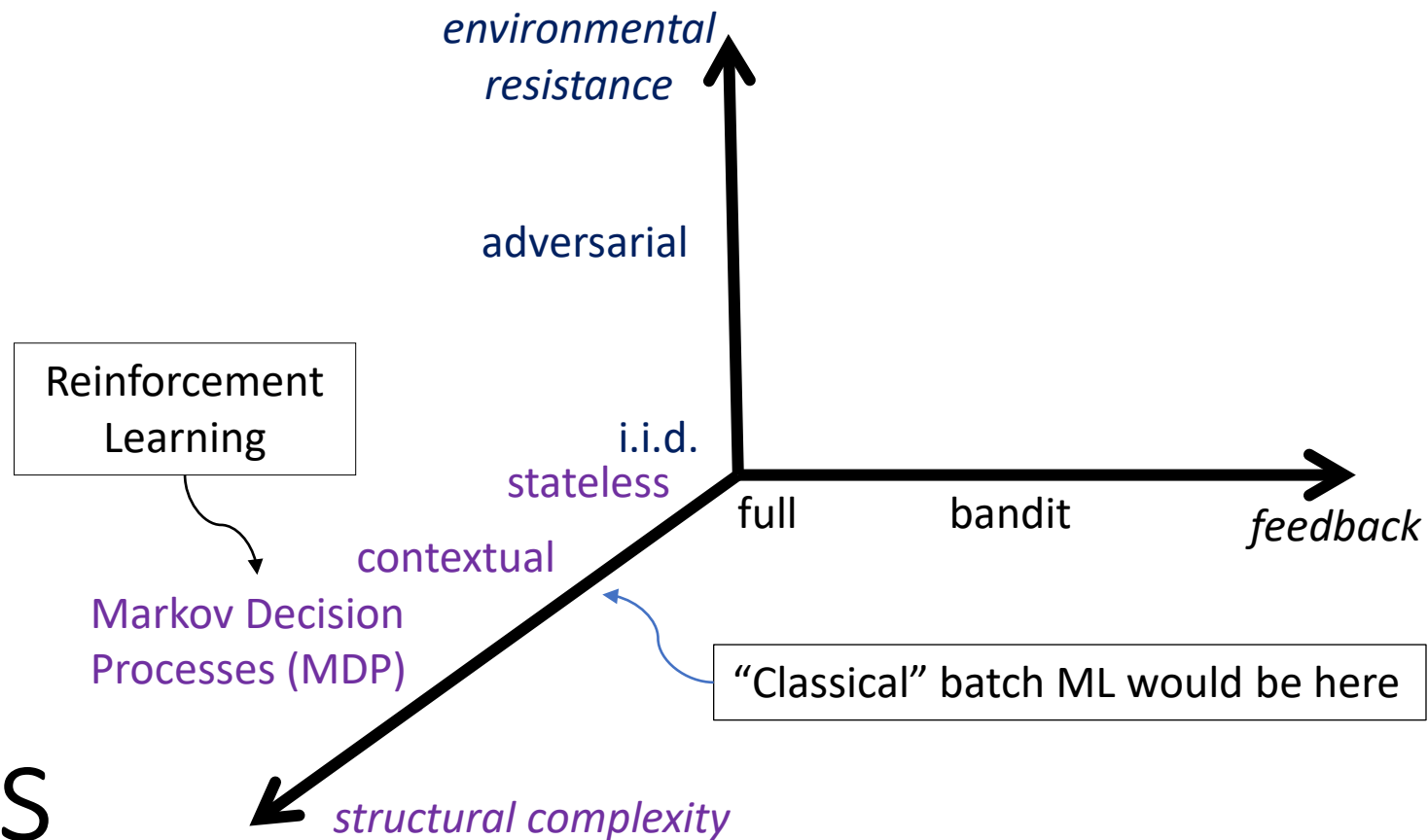
Stochastic Regime

$$K = 8, \Delta = 0.125$$



Stochastically Constrained
Adversarial Regime

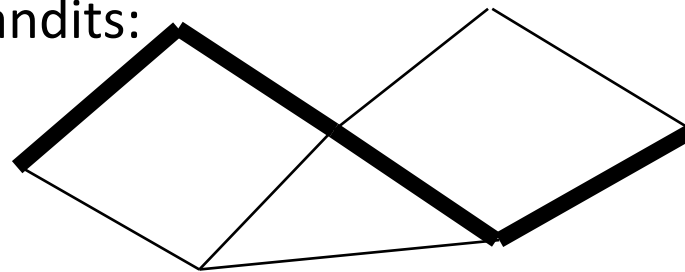
Other Settings



Structure forms: (Generalized) Linear Bandits

Linear Bandits:

- $r_t = \langle \bar{A}_t, \bar{\theta}_* \rangle + \xi_t$
- $\bar{A}_t \in \mathcal{D}_t$
- Special cases:
 - $\mathcal{D} = \{(1, 0, \dots, 0), \dots, (0, \dots, 0, 1)\}$ - multiarmed bandits
 - $\mathcal{D}_t = \{\phi(c_t, a) : a \in \{1, \dots, K\}\}$ - contextual bandits
 - Combinatorial (semi-)bandits:
 - Cascading bandits



Generalized Linear Bandits:

- $r_t = f(\langle \bar{A}_t, \bar{\theta}_* \rangle) + \xi_t$

Feedback forms

- From full to limited: paid observations, decoupled exploration, graph feedback, ...
- Dueling Bandits
 - Relative comparison of pairs arms, but not their true value
 - Would you like fish or chicken?
 - Very useful for implicit information collection from user feedback
- Ranking
 - Selection from a ranked list
- Partial Monitoring
 - Separation between observations and losses
 - Example: dynamic pricing

Environment forms

- Contaminated stochastic
- Stochastically constrained adversarial

Bandit variations

- Bandits with switching costs
- Recharging/recovering bandits
- Rotting bandits
- Bandits with knapsacks
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Delayed feedback

Alternative objectives

- We have studied regret minimization
 - Cumulative loss of actions along the way
- Pure Exploration / Best arm identification / Experiment design
 - Find the best action as fast as possible
 - Losses along the way are not counted

Summary

- An infinite world of exciting problem formulations